



MATERIAL SAFETY DATA SHEET

P3-TSUNAMI 100

Section: 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : P3-TSUNAMI 100

Other means of identification : Not applicable.

Recommended use : Biocide

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 0.06 %

Company : Ecolab Food Safety & Hygiene Solutions Pvt. Ltd
"215-Atrium" A WING, UNIT # 808,
Next to Courtyard by Marriott,
Andheri Kurla Road, Andheri (E)
Mumbai - 400 059, India
Toll free number: 1800 209 2530

Issuing date : 10/13/2016

Section: 2. COMPOSITION/INFORMATION ON INGREDIENTS

Product AS SOLD

Pure substance/mixture : Mixture

Chemical Name	CAS-No.	Concentration: (%)
Acetic acid	64-19-7	10 - 30
Peroxyacetic acid	79-21-0	10 - 30

Product AT USE DILUTION

No hazardous ingredients

Section: 3. HAZARDS IDENTIFICATION

Product AS SOLD

Physical and chemical hazards : May cause fire.

Potential Health Effects

Eyes : Causes serious eye damage.

Skin : Causes severe skin burns. Harmful in contact with skin.

Ingestion : Causes digestive tract burns. Harmful if swallowed.

Inhalation : May cause respiratory tract irritation. Harmful if inhaled.

Chronic Exposure : Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Physical and chemical hazards : Not flammable or combustible.

Potential Health Effects

MATERIAL SAFETY DATA SHEET

P3-TSUNAMI 100

Eyes	: Health injuries are not known or expected under normal use.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Section: 4. FIRST AID MEASURES

Product AS SOLD

In case of eye contact	: Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.
In case of skin contact	: Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.
If swallowed	: Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.
If inhaled	: Remove to fresh air. Treat symptomatically. Get medical attention.
Protection of first-aiders	: If potential for exposure exists refer to Section 8 for specific personal protective equipment.
Notes to physician	: Treat symptomatically.
Most important symptoms and effects, both acute and delayed	: See Section 11 for more detailed information on health effects and symptoms.

Product AT USE DILUTION

In case of eye contact	: Rinse with plenty of water.
In case of skin contact	: Rinse with plenty of water.
If swallowed	: Rinse mouth with water.
If inhaled	: Get medical attention if symptoms occur.

Section: 5. FIREFIGHTING MEASURES

Product AS SOLD

Suitable extinguishing media	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	: None known.
Specific hazards during firefighting	: Fire Hazard Keep away from heat and sources of ignition. Flash back possible over considerable distance. Special protective equipment for firefighters

MATERIAL SAFETY DATA SHEET

P3-TSUNAMI 100

Oxidizer; material is an oxidizer which may readily react with other materials, especially upon heating.

- Hazardous combustion products : Decomposition products may include the following materials:
Carbon oxides
Oxides of phosphorus
- Special protective equipment for firefighters : In case of fire, wear a full face positive-pressure self contained breathing apparatus and protective suit.
- Specific extinguishing methods : Use water spray to cool unopened containers. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

Section: 6. ACCIDENTAL RELEASE MEASURES

Product AS SOLD

- Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Remove all sources of ignition. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.
- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Eliminate all ignition sources if safe to do so. Stop leak if safe to do so. Never soak up spilled or leaked acids and bases with sawdust, wood chips or similar materials. Isolate the waste do not allow it to come into contact with incompatible materials. For small spills contain with sand or vermiculite and dilute the contained product at least 10 times with water. Transfer to an open topped container and remove to a safe place for neutralization* / disposal. For large spills contain spill and evacuate the area, leave until the reaction subsides, then collect up for disposal. Obtain consent from the local water company / authority if considering discharge to sewer. *NEUTRALIZATION : once diluted, neutralize with a suitable alkali such as sodium bicarbonate.

Product AT USE DILUTION

- Personal precautions, protective equipment and emergency procedures : Refer to protective measures listed in sections 7 and 8.
- Environmental precautions : No special environmental precautions required.
- Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

Section: 7. HANDLING AND STORAGE

Product AS SOLD

- Advice on safe handling : Do not ingest. Do not breathe dust/fume/gas/mist/vapours/spray. Use only with adequate ventilation. Keep away from fire, sparks and

MATERIAL SAFETY DATA SHEET

P3-TSUNAMI 100

heated surfaces. Take necessary action to avoid static electricity discharge (which might cause ignition of organic vapours). Wash hands thoroughly after handling. Do not get in eyes, on skin, or on clothing. Do not mix with bleach or other chlorinated products – will cause chlorine gas.

Conditions for safe storage : Keep away from heat and sources of ignition. Keep in a cool, well-ventilated place. Keep away from oxidizing agents. Keep away from reducing agents. Keep away from strong bases. Keep away from combustible material. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers. Pressure bursts may occur due to gas evolution if the container is not adequately vented.

Storage temperature : -20 °C to 30 °C

Packaging material : Suitable material: Plastic material, including expanded plastics material
Unsuitable material: Aluminium, Mild steel

Product AT USE DILUTION

Advice on safe handling : Wash hands after handling. For personal protection see section 8.

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Components with workplace control parameters

Components	CAS-No.	Form of exposure	Permissible concentration	Basis
Acetic acid	64-19-7	TWA	10 ppm 25 mg/m3	IN OEL
		STEL	15 ppm 37 mg/m3	IN OEL
Hydrogen peroxide	7722-84-1	TWA	1 ppm 1.5 mg/m3	IN OEL

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

Personal protective equipment

Eye protection : Safety goggles
Face-shield

Hand protection : Wear the following personal protective equipment:
Standard glove type.
PVC
Laminate film
Unsupported neoprene
Natural rubber
Neoprene/natural rubber blend
Nitrile
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

MATERIAL SAFETY DATA SHEET

P3-TSUNAMI 100

Wear the following personal protective equipment:

Standard glove type.

Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Appearance	: liquid	liquid
Colour	: colourless	colourless
Odour	: pungent	vinegar-like
pH	: 0.5 - 1.5, 100 %	4.0 - 7.5
Flash point	: 72 °C closed cup	
Odour Threshold	: no data available	
Melting point/freezing point	: no data available	
Initial boiling point and boiling range	: > 100 °C	
Evaporation rate	: no data available	
Flammability (solid, gas)	: no data available	
Upper explosion limit	: no data available	
Lower explosion limit	: no data available	
Vapour pressure	: no data available	
Relative vapour density	: no data available	
Relative density	: 1.13 - 1.15	
Water solubility	: soluble	

MATERIAL SAFETY DATA SHEET

P3-TSUNAMI 100

Solubility in other solvents	: no data available
Partition coefficient: n-octanol/water	: no data available
Auto-ignition temperature	: no data available
Thermal decomposition	: no data available
Viscosity, kinematic	: no data available
Explosive properties	: no data available
Oxidizing properties	: YesOxidising solid - Type 3.
Molecular weight	: no data available
VOC	: no data available

Section: 10. STABILITY AND REACTIVITY

Product AS SOLD

Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: Do not mix with bleach or other chlorinated products – will cause chlorine gas.
Conditions to avoid	: Heat, flames and sparks. Direct sources of heat. Exposure to sunlight.
Incompatible materials	: Bases Metals Organic materials
Hazardous decomposition products	: Decomposition products may include the following materials: Carbon oxides Oxides of phosphorus

Section: 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Product AS SOLD

Eyes	: Causes serious eye damage.
Skin	: Causes severe skin burns. Harmful in contact with skin.
Ingestion	: Causes digestive tract burns. Harmful if swallowed.
Inhalation	: May cause respiratory tract irritation. Harmful if inhaled.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Eyes	: Health injuries are not known or expected under normal use.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.

MATERIAL SAFETY DATA SHEET

P3-TSUNAMI 100

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

Experience with human exposure

Product AS SOLD

Eye contact : Redness, Pain, Corrosion

Skin contact : Redness, Pain, Corrosion

Ingestion : Corrosion, Abdominal pain

Inhalation : Respiratory irritation, Cough

Product AT USE DILUTION

Eye contact : No symptoms known or expected.

Skin contact : No symptoms known or expected.

Ingestion : No symptoms known or expected.

Inhalation : No symptoms known or expected.

Toxicity

Product AS SOLD

Product

Acute oral toxicity : Acute toxicity estimate : 1,363 mg/kg

Acute inhalation toxicity : 4 h Acute toxicity estimate : 24.98 mg/l

Acute dermal toxicity : Acute toxicity estimate : 2,609 mg/kg

Skin corrosion/irritation : no data available

Serious eye damage/eye irritation : no data available

Respiratory or skin sensitization : no data available

Carcinogenicity : no data available

Reproductive effects : no data available

Germ cell mutagenicity : no data available

Teratogenicity : no data available

STOT - single exposure : no data available

STOT - repeated exposure : no data available

Aspiration toxicity : no data available

Section: 12. ECOLOGICAL INFORMATION

Product AS SOLD

Ecotoxicity

Environmental Effects : Toxic to aquatic life with long lasting effects.

Product

MATERIAL SAFETY DATA SHEET

P3-TSUNAMI 100

Toxicity to fish : no data available

Toxicity to daphnia and other aquatic invertebrates : no data available

Toxicity to algae : no data available

Components

Toxicity to fish : Acetic acid
96 h LC50: 75 mg/l

Peroxyacetic acid
96 h LC50: 0.8 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates : Peroxyacetic acid
48 h EC50: 0.73 mg/l

Components

Toxicity to algae : Peroxyacetic acid
72 h EC50: 0.7 mg/l

Persistence and degradability

Product AS SOLD

Readily biodegradable.

Bioaccumulative potential

no data available

Mobility in soil

no data available

Other adverse effects

no data available

Section: 13. DISPOSAL CONSIDERATIONS

Product AS SOLD

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

The product should not be allowed to enter drains, water courses or the soil.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Product AT USE DILUTION

Disposal methods : Diluted product can be flushed to sanitary sewer.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

Section: 14. TRANSPORT INFORMATION

MATERIAL SAFETY DATA SHEET

P3-TSUNAMI 100

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport

UN number : 3098
Description of the goods : OXIDIZING LIQUID, CORROSIVE, N.O.S.
(Peroxyacetic acid, acetic acid, Hydrogen peroxide)
Class : 5.1 (8)
Packing group : III
Environmentally hazardous : No

Air transport (IATA)

UN number : 3098
Description of the goods : Oxidizing liquid, corrosive, n.o.s.
(Hydrogen peroxide, Peroxyacetic acid, acetic acid)
Class : 5.1 (8)
Packing group : III
Environmentally hazardous : Yes

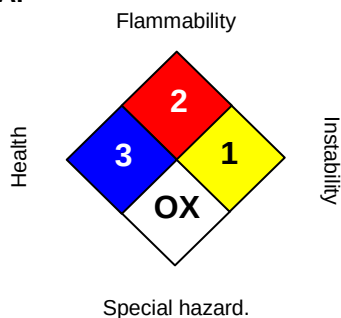
Sea transport (IMDG/IMO)

UN number : 3098
Description of the goods : OXIDIZING LIQUID, CORROSIVE, N.O.S.
(Hydrogen peroxide, Peroxyacetic acid, acetic acid)
Class : 5.1 (8)
Packing group : III
Marine pollutant : Yes

Section: 15. REGULATORY INFORMATION

Product AS SOLD

NFPA:



Symbol(s) : O - OXIDIZING
C - CORROSIVE
N - DANGEROUS FOR THE ENVIRONMENT

R-phrases(s) : R07 - May cause fire.
R20/21/22 - Harmful by inhalation, in contact with skin and if swallowed.
R35 - Causes severe burns.
R37 - Irritating to respiratory system.
R51/53 - Toxic to aquatic organisms, may cause long-term adverse

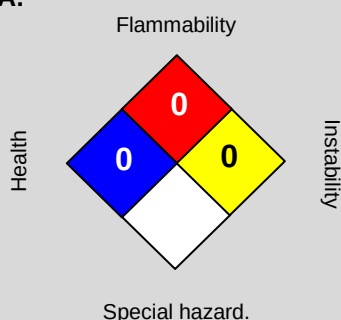
MATERIAL SAFETY DATA SHEET

P3-TSUNAMI 100

effects in the aquatic environment.

S-phrase(s) : S26 - In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
S36/37/39 - Wear suitable protective clothing, gloves and eye/face protection.
S45 - In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible).
S57 - Use appropriate container to avoid environmental contamination.
S60 - This material and/or its container must be disposed of as hazardous waste.

Product AT USE DILUTION NFPA:



Not a hazardous substance or mixture.

The components of this product are reported in the following inventories:

United States TSCA Inventory :
not determined

Canadian Domestic Substances List (DSL) :
This product contains the following components that are not on the Canadian DSL nor NDSL.

Australia. Industrial Chemical (Notification and Assessment) Act :
not determined

New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand :
not determined

Japan. ENCS - Existing and New Chemical Substances Inventory :
not determined

Japan. ISHL - Inventory of Chemical Substances (METI) :
not determined

Korea. Korean Existing Chemicals Inventory (KECI) :
not determined

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :
not determined

China Inventory of Existing Chemical Substances :
not determined

MATERIAL SAFETY DATA SHEET

P3-TSUNAMI 100

Section: 16. OTHER INFORMATION

Issuing date : 10/13/2016
version : 1.1
Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the MSDS.

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.